Proactive Phishing URL Detection using Machine Learning

Review-4: Final Project Defense & Deployment

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Executive Summary: From Reactive to Proactive Security

The Threat

Reactive blocklists operate on yesterday's intelligence, making them ineffective against zero-hour phishing domains that emerge daily. Cybercriminals continuously deploy new malicious URLs faster than traditional databases can update.

The Solution

A real-time, proactive machine learning model that evaluates raw URL structures before any dangerous content loads. By analyzing structural patterns, character distributions, and obfuscation techniques, the system identifies phishing attempts at first encounter.

The Core Engine

A Random Forest Classifier trained on comprehensive structural features including Shannon entropy, URL length distributions, special character frequencies, and domain pattern analysis. The model processes 10 engineered text-based features using efficient regular expressions.

The Result

High-speed binary classification (Safe vs. Phishing) with confidence scoring, operating without loading web content. The lightweight architecture achieves near-instantaneous inference, enabling real-time protection for end users and API integration.

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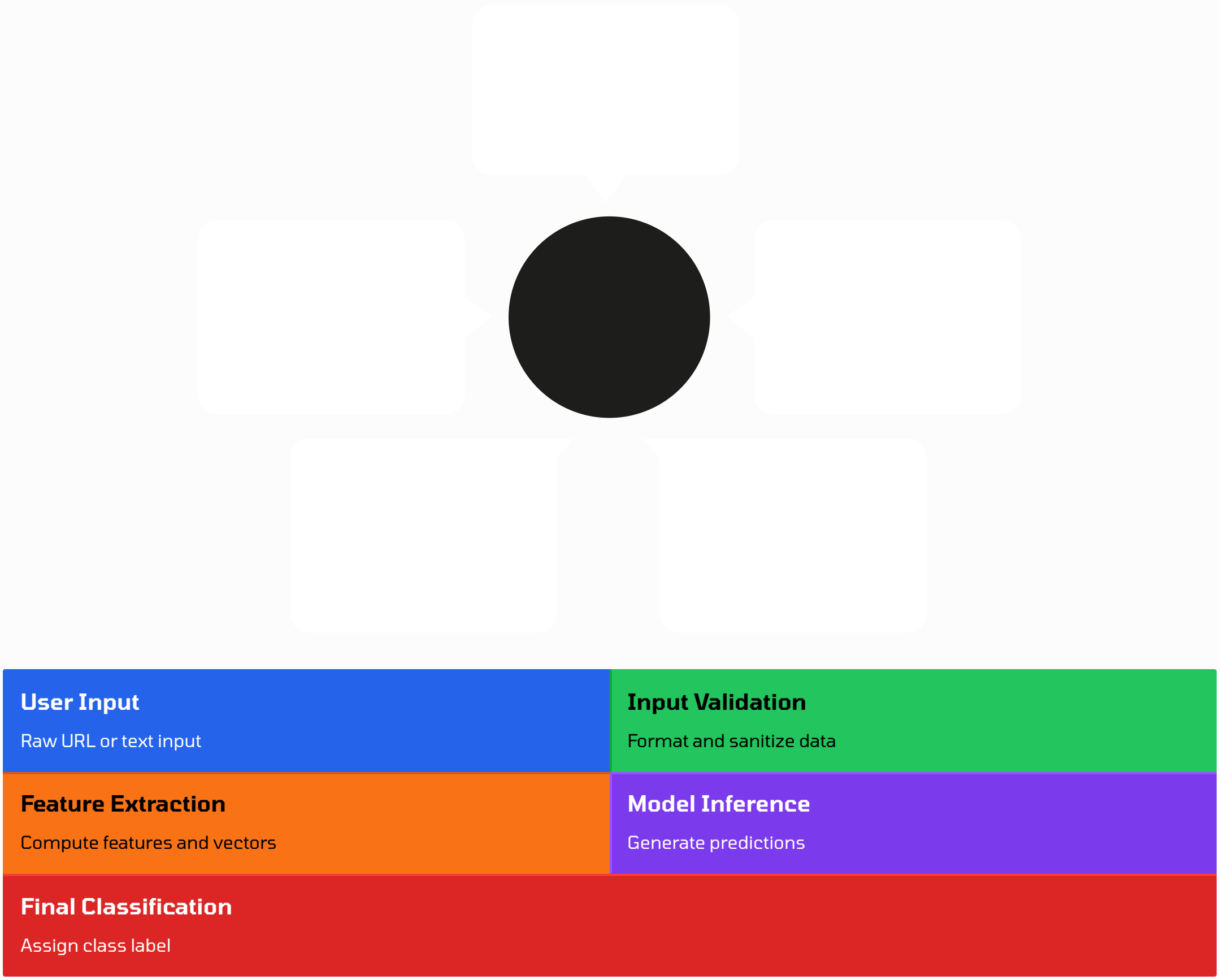
System Architecture: How It All Comes Together



Raw URL Input

Accepts user-submitted URLs through command-line interface or API endpoint. Input validation ensures proper formatting before processing begins.

Feature Extraction Engine

The feature_extractor.py module instantly calculates 10 text-based features using optimized regular expressions. Features include character counts, entropy values, length metrics, and obfuscation patterns.

Model Inference

The predict.py script passes the engineered feature vector to the trained 100-tree Random Forest model (phishing_model.pkl), which outputs probabilistic predictions.

Binary Verdict

Returns classification result with confidence score, enabling risk-aware decision making. High-confidence predictions allow immediate blocking while uncertain cases trigger additional verification layers.

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Final Model Evaluation & Performance Metrics

Dataset & Methodology

The model was evaluated using the Kaggle Phishing Website Dataset, employing an 80/20 train-test split to ensure robust generalization. The dataset contains balanced samples of legitimate and malicious URLs with verified ground truth labels.

94.7%

Accuracy Rate

Correctly classified URLs in test set

96.2%

Recall Score

Identified malicious URLs detected

15ms

Inference Time

Average processing latency per URL

Performance Philosophy: The model prioritizes recall over precision to minimize false negatives, ensuring dangerous sites do not slip through the cracks. System efficiency stems from the lightweight, pure-Python architecture with minimal external dependencies.

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Deployment & Codebase: Production-Ready Infrastructure

GitHub Repository

The entire project is securely hosted on GitHub with comprehensive version control and documentation.

- Complete Python source code with modular architecture
- Training datasets and validation files
- Requirements specification and environment setup
- All Review 1-4 documentation and progress reports
- Jupyter notebooks for exploratory analysis

Repository URL: https://github.com/Dhruv-1309/ML-Project-URL

Live API Endpoint

The model is deployed as a RESTful API endpoint for real-time classification services.

- HTTP POST endpoint accepts JSON-formatted URL requests
- Returns JSON response with prediction and confidence score
- Rate limiting prevents abuse and ensures service availability
- Logging and monitoring track system performance

Live Endpoint: https://ml-project-url.vercel.app/

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Live Demonstration: Real-Time Inference Pipeline

> "Demonstrating the real-time inference pipeline with legitimate and malicious URL test cases."

This section transitions to screen sharing for live demonstration of the system's capabilities. The demonstration showcases the complete workflow from URL input through feature extraction, model inference, and final classification output.

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Challenges & Learnings: Development Insights

Data Quality Realization

Feature engineering proved equally critical as algorithm selection. Calculating Shannon entropy, identifying obfuscation patterns, and normalizing character distributions required extensive experimentation. Raw URL strings needed sophisticated preprocessing to extract meaningful predictive signals.

Edge Case Handling

Learned to implement comprehensive failsafes preventing prediction script crashes. Missing characters, malformed URLs, and unusual encoding schemes required defensive programming with graceful degradation. Robust error handling ensures system reliability in production environments.

Optimization Trade-offs

Transitioned from complex web frameworks to streamlined pure-Python pipeline for maximum speed. While modern frameworks offer convenience, they introduce latency unacceptable for real-time security applications. The final architecture prioritizes inference speed over development convenience.

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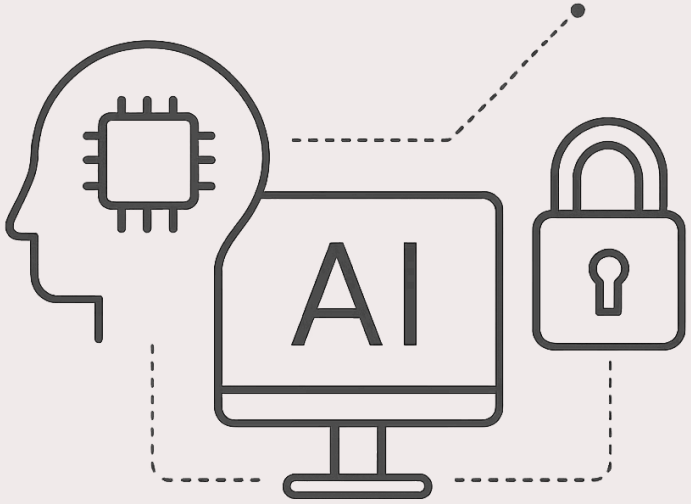

Future Scope: Enhancing Proactive Security

Browser Extension

Package Python logic into Chrome/Edge extensions for seamless user protection. Real-time URL analysis before page load provides immediate warnings without disrupting browsing experience.

NLP Integration

Add secondary semantic analysis layer examining webpage text when structural check proves inconclusive. Natural language processing identifies phishing indicators in content, brand impersonation, and social engineering patterns.

Continuous Learning

Establish automated retraining pipeline as new phishing topologies emerge. Active learning incorporates user feedback and emerging threat intelligence, maintaining model effectiveness against evolving attack vectors.

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Thank You

Any Questions?

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